C12.51

DATA SET
DS1201

21. **How Long Is It?** How good are you at estimating? To test a subject's ability to estimate sizes, he was shown 10 different objects and asked to estimate their length or diameter. The object was then measured, and the results were recorded in the table below.

Object	Estimated (centimeters) x	Actual (centimeters) y
Pencil	17.7	15.2
Dinner plate	24.1	26.0
Book 1	19.0	17.1
Cell phone	10.1	10.7
Photograph	36.8	40.0
Toy	9.5	12.7
Belt	106.6	105.4
Clothespin	6.9	9.5
Book 2	25.4	23.4
Calculator	8.8	12.0

a. Find the least-squares regression line for predicting the actual measurement as a function of the estimated measurement.

b. Plot the points and the fitted line. Does the assumption of a linear relationship appear to be reasonable?

a. $\hat{y} = a + bx$

$b = \frac{S_{xy}}{S_{xx}} = \frac{7717.13}{7918.2} = 0.975$ $\quad a = \bar{y} - b\bar{x} = 27.2 - 0.975 \times 26.49 = 1.372$

$S_{xy} = r \cdot (n-1) \cdot s_x \cdot s_y = 0.997 \times 28.99 \times 29.66 \times 9 = 7717.13$

$S_{xx} = (n-1) s_x^2 = 9 \times 879.8 = 7918.2$

$s_x^2 = 879.8$

$s_y = 28.99$ $\quad s_x = 29.66$

$r = 0.997$

$\Rightarrow \hat{y} = 1.372 + 0.975x$